

ZoneA $\leftarrow 0$, ZoneB $\leftarrow 0$, ZoneC $\leftarrow 0$

MaxA $\leftarrow 20$, MaxB $\leftarrow 40$, MaxC $\leftarrow 15$

Accepted $\leftarrow 0$, Rejected $\leftarrow 0$

Car count $\leftarrow 0$, Bikecount $\leftarrow 0$, vancount $\leftarrow 0$

Input N

For i $\leftarrow 1$ TO N

Print "Enter vehicle type (C/B/V):"

Input vehicle

Repeat

If vehicle $\neq$ 'C' AND vehicle $\neq$ 'B' AND vehicle $\neq$ 'V' then

Print "Invalid vehicle type."

Input vehicle

End If

Until vehicle == 'C' OR vehicle == 'B' OR vehicle == 'V'

Print "Enter user category (F/S/G):"

Input Category

Repeat

If category $\neq$ 'F' AND category $\neq$ 'S' AND category $\neq$ 'G' then

Print "Invalid category"

End If

Until category == 'F' OR category == 'S' OR category == 'G'.

Print "Enter parking permit (Y/N) : "

Input Permit

Repeat

If permit != 'Y' AND permit != 'N' then

Print "Invalid permit. Enter again"

Input Permit

End If.

Until permit == 'Y' OR permit == 'N'.

emergency ← 'N'

If permit == 'N' then

Print "Is this emergency vehicle? Y/N : "

Input emergency

Repeat

If emergency != 'Y' AND emergency != 'N' then

Print "Invalid value. Enter again"

Input emergency.

End If.

Until emergency == 'Y' OR emergency == 'N'.

End If

assigned ← 'N' , zone ← 'X' , spaces ← 1.

If Vehicle == 'V' then

spaces = 2.

Else space = 1.

End If

If permit == 'N' AND emergency == 'N' then
Print "Rejected"

Rejected $\leftarrow$ Rejected + 1.

Else

If category == 'F' Then

If zoneA + space $\leq$ maxA then

zoneA $\leftarrow$ zoneA + spaces

zone $\leftarrow$ 'A'

assigned $\leftarrow$ 'Y'

Else Print "Rejected"

rejected $\leftarrow$ rejected + 1

End If

Else If category == 'S' then

If vehicle == 'V' then

If zoneC + 2 $\leq$ maxC then

zoneC $\leftarrow$ zoneC + 2

zone $\leftarrow$ 'C'

assigned = 'Y'

Else Print "Rejected"

rejected $\leftarrow$ rejected + 1

End If

Else If zoneB + 1 $\leq$ maxB then

zoneB $\leftarrow$ zoneB + 1

zone $\leftarrow$ 'B'

assigned $\leftarrow$ 'Y'

Else Print "Rejected"

Rejected $\leftarrow$ Rejected + 1

End If

End If End If

If assigned == 'Y' then

accepted $\leftarrow$ accepted + 1

If vehicle == 'C' then

carcount $\leftarrow$ carcount + 1

Else If vehicle == 'B' then

bikecount $\leftarrow$ bikecount + 1

Else If vehicle == 'V' then

vancount $\leftarrow$ vancount + 1

End If

Print "vehicle parked"

Print "assigned zone", zone

If zone == 'A' then

Print "Remaining capacity", maxA - zoneA

Else If zone == 'B' then

Print "Remaining capacity", maxB - zoneB

Else Print "Remaining capacity", maxC - zoneC.

End If

End If

End If

End For

Print N

Print accepted

Print rejected

Print car count

Print bikecount

Print van count

Print " Occupied zone A ", zoneA
Print " Zone A remaining capacity ", $\text{max A} - \text{zoneA}$
Print " Occupied zone B ", zoneB
Print " zone B remaining capacity ", $\text{max B} - \text{zoneB}$
Print " Occupied zone C ", zoneC
Print " zone C remaining capacity ", $\text{max C} - \text{zoneC}$

If zoneA > zoneB AND zoneA > zoneC then,

Print " Highest occupied zone is A "

Else If zoneB > zoneA AND zoneB > zoneC then,

Print " Highest occupied zone is B "

Else print " Highest occupied zone is C "

End If.

If zoneA == maxA AND zoneB == maxB AND zoneC == maxC
then print " Entire parking is full ".